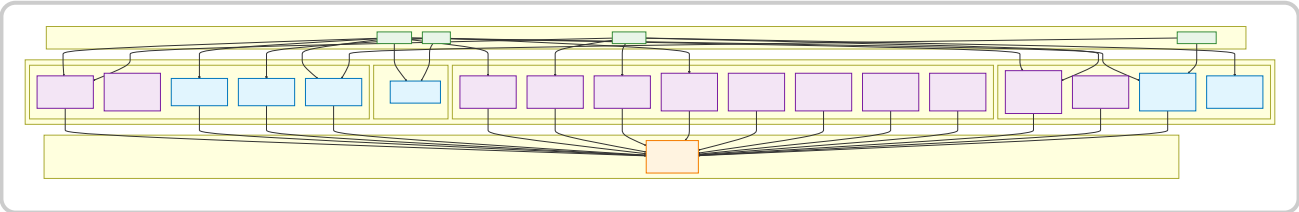


BK8520 Electronic Load REST API Flow Diagram

This Mermaid diagram illustrates the POST and GET API endpoints for the BK8520 Electronic Load Control system.

API Architecture Overview



API Endpoint Details

GET Endpoints (Data Retrieval)

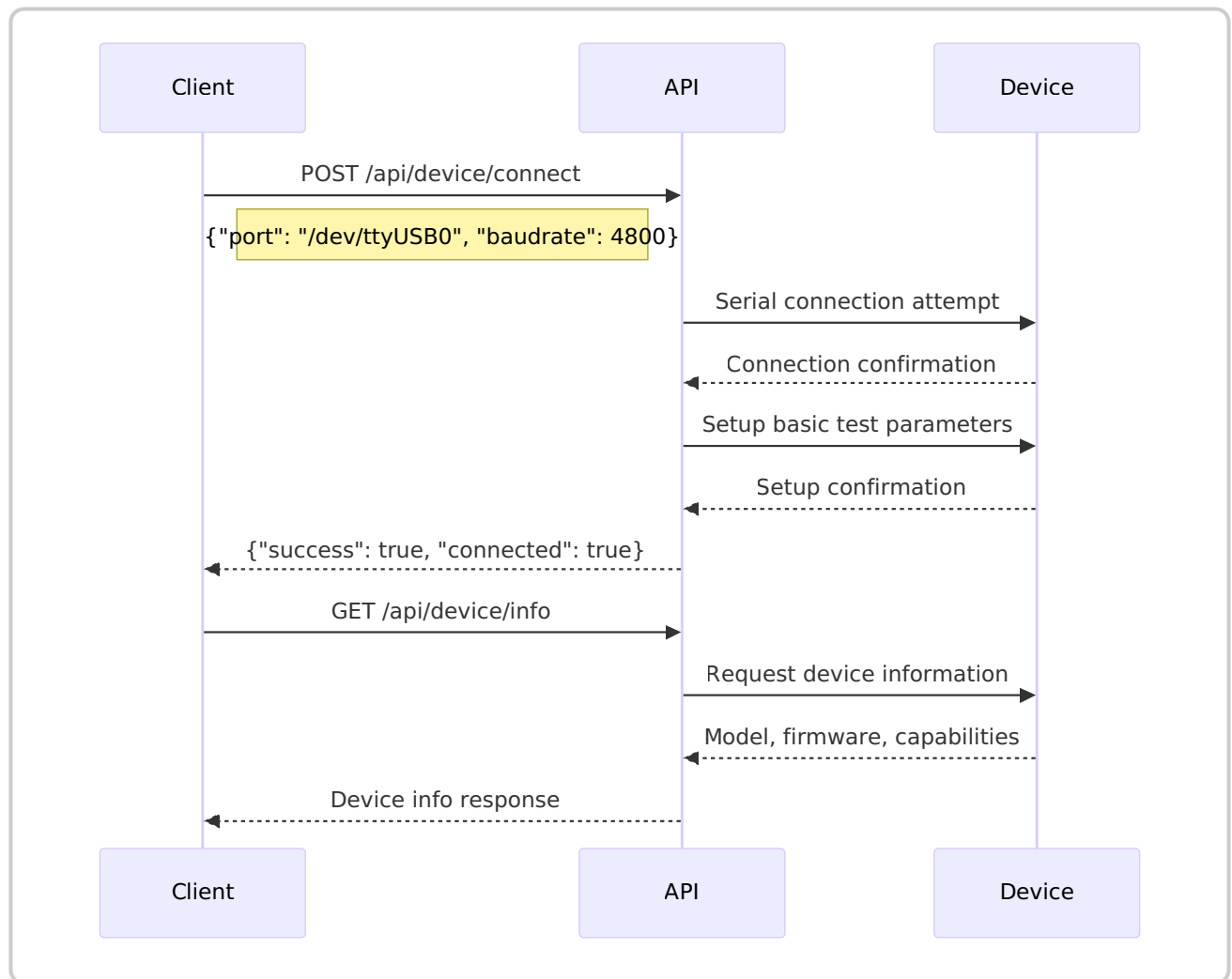
Endpoint	Purpose	Response Data
GET /api/health	Service health check	Status and version info
GET /api/device/status	Connection and operational status	Connection state, device status
GET /api/device/info	Device information	Model, firmware, capabilities
GET /api/device/readings	Real-time measurements	Voltage, current, power values
GET /api/battery-test/status	Active test status	Test progress, measurements
GET /api/battery-test/results	Complete test results	Full test data and summary

POST Endpoints (Control Operations)

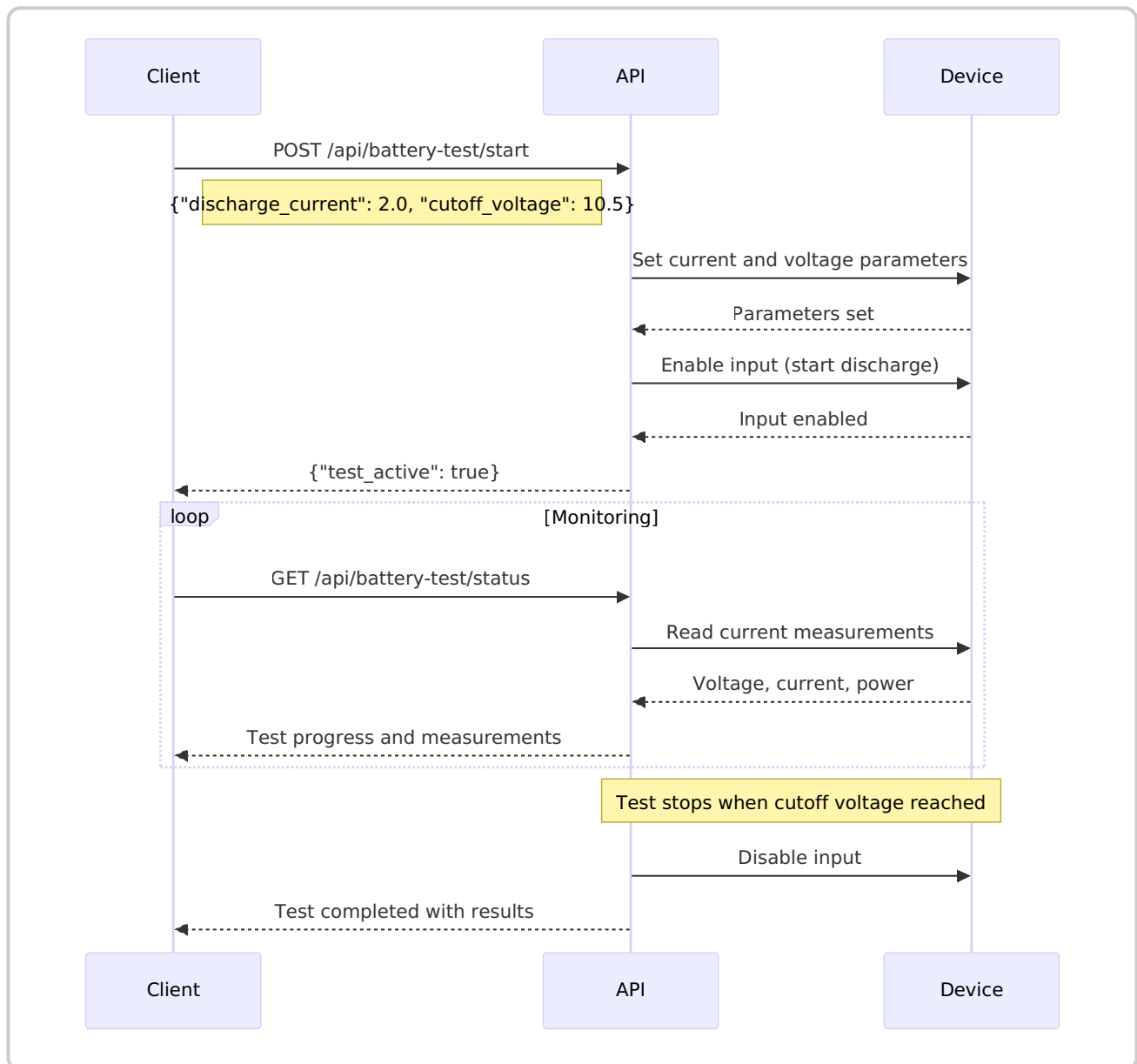
Endpoint	Purpose	Parameters
POST /api/device/connect	Connect to device	Port, baudrate, timeout
POST /api/device/disconnect	Disconnect device	None
POST /api/device/input	Control input on/off	enabled (boolean)
POST /api/device/current	Set discharge current	current (amperes)
POST /api/device/voltage	Set cutoff voltage	voltage (volts)
POST /api/device/mode	Set operation mode	mode (0=CC, 1=CV, 2=CW, 3=CR)
POST /api/device/max-current	Set current limit	current (amperes)
POST /api/device/max-power	Set power limit	power (watts)
POST /api/device/max-voltage	Set voltage limit	voltage (volts)
POST /api/device/setup-discharge	Setup CC discharge	current (amperes)
POST /api/battery-test/start	Start battery test	discharge_current, cutoff_voltage, max_time_hours
POST /api/battery-test/stop	Stop battery test	None

Data Flow Examples

Device Connection Flow



Battery Test Flow



System Architecture

The API server acts as a bridge between client applications and the physical BK8520 device:

- **FastAPI Server:** Handles HTTP requests and provides OpenAPI documentation
- **Serial Communication:** Direct communication with BK8520 via RS232/USB
- **Real-time Monitoring:** Continuous measurement reading during tests
- **Data Persistence:** Test results saved to JSON files
- **Error Handling:** Comprehensive error reporting and device safety checks

Base URL: <http://10.100.10.190:8000>